Abhi Jain

B.Tech Artificial Intelligence Student | Machine Learning & Data Science Internship Seeker

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LinkedIn | GitHub | HuggingFace | Portfolio

PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence student (2024–2028) seeking a Machine Learning / Data Science internship. Built and deployed supervised, unsupervised, reinforcement learning, and NLP projects in Python (Scikit-learn, PyTorch, Hugging Face), from EDA and feature engineering to live deployment with FastAPI, Docker, and GitHub Actions. VoidHacks8 hackathon finalist (top 64 of 850 applications): built the anomaly-detection layer (Isolation Forest) of a mule-account detection platform analyzing 2M transactions.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: Supervised Learning, Unsupervised Learning, Logistic Regression, KNN, Decision Trees, Random Forest, SVM, K-Means Clustering, PCA, Anomaly Detection (Isolation Forest), Cross-Validation, Model Evaluation

Deep Learning & NLP: PyTorch, Deep Q-Networks (DQN), Reinforcement Learning, Transformers (T5 fine-tuning), Text Summarization, Hugging Face

Data Science: EDA, Feature Engineering, Data Cleaning, Data Preprocessing, Dimensionality Reduction, Data Visualization, Customer Segmentation, Fraud Detection

Libraries & Tools: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Gymnasium, Gradio, Git, GitHub, GitHub Actions (CI/CD), Docker, FastAPI (REST API), Hugging Face Spaces, Streamlit, Jupyter Notebook

Currently Learning: Computer Vision, Power BI, Hyperparameter Tuning, Advanced Regression Models

PROJECTS

Abhedya Chakra — Mule-Account Detection & Fund Tracing (Team Project, VoidHacks8) GitHub

- Contributed to a team-built investigation platform (FastAPI + React) on the VoidHacks8 Mule Dataset of **2,000,000 transactions**, featuring graph-based layer analysis, chronological fund tracing, and network visualization.
- Owned the risk-screening layer: implemented the final-risk-percentile rule (80th percentile or higher) and an **Isolation Forest** anomaly detection model on account risk and graph-behavior features.
- Narrowed **4,968 rule-based candidate accounts to 1,227** (**75.3%** smaller review pool) through the hybrid rule + machine learning screen, prioritizing leads for investigators.
- ***Tech:*** *Python, Scikit-learn, Pandas, Isolation Forest, Anomaly Detection*

Flappy Bird AI Agent — Deep Q-Network (DQN) Reinforcement Learning GitHub

- Built a reinforcement learning agent from scratch using DQN that learns Flappy Bird through trial and error, reaching a best episode reward of **57.1** (~50 pipes crossed) over **99,500+** training episodes.
- Implemented Experience Replay, a Target Network, and Epsilon-Greedy exploration; designed a 12-128-128-2 Q-network in PyTorch with a custom training loop and checkpoint auto-resume.
- ***Tech:*** *Python, PyTorch, Gymnasium, Reinforcement Learning, DQN*

Customer Market Segmentation using PCA & K-Means Live Demo on Streamlit

- Designed an unsupervised machine learning pipeline on real-world market data; applied PCA for dimensionality reduction and selected optimal clusters using the elbow method and silhouette analysis.
- Visualized customer segments in 2D and 3D scatter plots to extract actionable business insights; deployed as an interactive Streamlit app.
- ***Tech:*** *Python, Pandas, NumPy, Scikit-learn, PCA, K-Means, Matplotlib, Seaborn, Streamlit*

Briefly — Text Summarizer (Fine-tuned T5) GitHub

- Fine-tuned a **T5** transformer model for NLP text summarization and served it through a **FastAPI** REST backend with a web interface.
- Containerized the app with **Docker** and deployed it on Hugging Face Spaces, hosting model weights in a Hugging Face model repository to keep the code repository lightweight.
- Automated deployment with a **GitHub Actions** CI/CD workflow that redeploys the Space on every push to the main branch.
- ***Tech:*** *Python, T5, Hugging Face, FastAPI, Docker, GitHub Actions*

EXPERIENCE

Hackathon Finalist, VoidHacks8 (36-Hour Hackathon) 1–3 Oct 2026

Organized by Madhya Pradesh Police Commissionerate & SVVV Indore

- Selected among the **top 64 teams** from **850 online applications** to compete in the 36-hour final round.
- Built the anomaly-detection layer (percentile rule + Isolation Forest) of Abhedya Chakra, cutting the suspicious-account candidate pool by 75.3% across 2M transactions.
- Collaborated in a team to deliver a working prototype (API + dashboard); open-sourced the code on GitHub with documented limitations (outputs are screening leads, not confirmed fraud).

EDUCATION

B.Tech in Artificial Intelligence, Sage University, Indore | 2024–2028 CGPA: 7.35/10

Higher Secondary (Class 12) | 2023 72%

Secondary (Class 10) | 2021 82%

Relevant Coursework: Machine Learning, Probability & Statistics, Data Structures, Linear Algebra

CERTIFICATIONS

Microsoft Azure Fundamentals (AZ-900) 2025

Apna College PRIME — AI & Machine Learning Batch (Ongoing) 2025